Canny Edge Detection Filter – Concept, Development, Use and Application

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1 1. Introduction

1.1 What is an Edge?

In a digital image, an edge is simply a boundary where pixel brightness changes rapidly over a short distance. For example, if you look at an image where dark sea water meets bright dry land, the line where the brightness changes from dark to light is an edge.

In real satellite images, edges are rarely sharp like a line drawn with a pen. Instead, they are usually "ramp edges," where the brightness changes gradually over two or three pixels because a single 10-meter pixel might contain both shallow water and wet sand.

Figure 1 shows the ideal shapes of edges: a step edge, a ramp edge and a roof edge, each with its appearance in an image and its brightness profile (Gonzalez & Woods, 2018). A fourth type, the line edge, is a roof edge whose base is only about one pixel wide (Gonzalez & Woods, 2018; Jain et al., 1995).

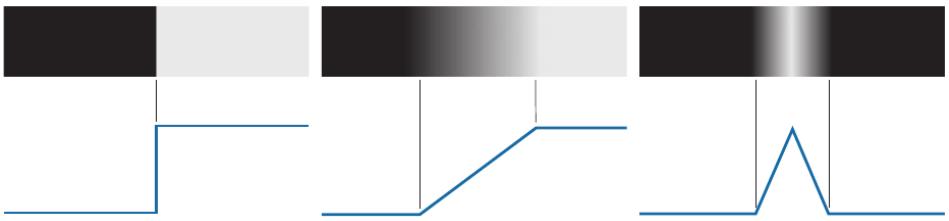


Figure 1. Models of a step, a ramp and a roof edge (top) and their brightness profiles (bottom). (Source: Gonzalez and Woods, 2018, Fig. 10.8.)

1.2 The Problem in Remote Sensing

Tracing water boundaries (such as coastlines, riverbanks, and lake perimeters) is an important task in remote sensing. It is used to monitor coastal erosion, update navigation maps, and track flood levels.

However, finding coastlines using simple brightness changes is surprisingly difficult:

- Roads, buildings, concrete docks, and agricultural field boundaries also have sharp brightness changes. A simple edge detector will mark all of them as edges.
- Ocean waves, boat wakes, and cloud shadows create false edges inside the water.
- Murky water with mud or sand has a very weak contrast with the land, making the true shore

difficult to detect.

1.3 Objective of this Term Paper

Since the Canny edge detector was not covered in our regular class lectures, this project has three main goals:

1. Explain clearly how the Canny edge detector works mathematically and conceptually.
2. Compare Canny with the edge filters taught in class (Roberts, Sobel, Prewitt, and Laplacian).
3. Test how well Canny extracts real water boundaries on Sentinel-2 satellite data from the UK Hydrographic Office (SWED dataset), and show how to make it run fast on large datasets.

2 2. Concept & Development: How Canny Works

2.1 How Spatial Filters Work

In class, we learned that edge filtering is done using a method called **convolution**. You take a tiny grid of numbers called a **kernel** (usually 3×3 pixels) and slide it across the image.

$$G_x \text{ kernel} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

At every pixel, you multiply the kernel values with the image pixel values underneath, and add them up.

Edge kernels are designed so that all their numbers add up to zero:

- If the filter is over a flat area (like open water where all pixels have a value of around 30), the positive and negative numbers cancel out, and the output is **0** (no edge).
- If the filter sits across an edge (dark water on the left, bright land on the right), the numbers do not cancel out, producing a **large positive or negative value** (strong edge).

By using two kernels: one for horizontal changes (Gx) and one for vertical changes (Gy).
We calculate two things at each pixel:

- Edge Strength (G) = sqrt(G_x^2+G_y^2)
- **Edge Direction (θ) = $\tan^{-1}$(Gy / Gx)**

2.2 Edge Filters Before Canny (Classroom Filters)

Between 1963 and 1980, several basic filters were invented:

1. **Roberts Cross (1963):** Uses tiny 2×2 diagonal kernels. Because it is so small, it is very fast, but it is extremely sensitive to image noise and speckle.
2. **Prewitt (1970):** Uses 3×3 kernels with equal weights. It calculates averages to reduce noise, but leaves wide, blurry edges.
3. **Sobel (1968):** Similar to Prewitt, but puts double weight on the center pixel ([−1,0,1;−2,0,2;−1,0,1]). This gives slightly better smoothing, but edges are still several pixels thick.

$$G_x \text{ kernel} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} G_y \text{ kernel} = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

4. **Laplacian:** Uses a second derivative (∇2). Instead of calculating slope, it looks for places where the second derivative crosses zero. Its major drawback is that it strongly amplifies noise, marking almost every tiny texture or speckle in the image.

5. **Laplacian of Gaussian (LoG, 1980):** Blurs the image with a Gaussian filter before taking the Laplacian. This reduces noise, but rounds off sharp corners and misses faint lines.

Table 1. Early edge filters.

Filter	Year	Main idea	Main problem
Roberts	1963	Very small 2 × 2 kernel	Easily disturbed by noise
Sobel	1968	3 × 3 kernel, extra weight on the centre row	Thick edges
Prewitt	1970	3 × 3 kernel, equal weights	Noise and thick edges
Laplacian	–	Looks for a change in all directions at once	Very sensitive to noise
LoG (Marr–Hildreth)	1980	Blur first, then Laplacian	Less noise, but corners get rounded

Main Problem with Class Filters: These filters only *highlight* bright gradient areas. They do not give you a clean, one-pixel-wide line, and they break into pieces whenever the edge gets slightly faint.

2.3 Canny's Solution (1986)

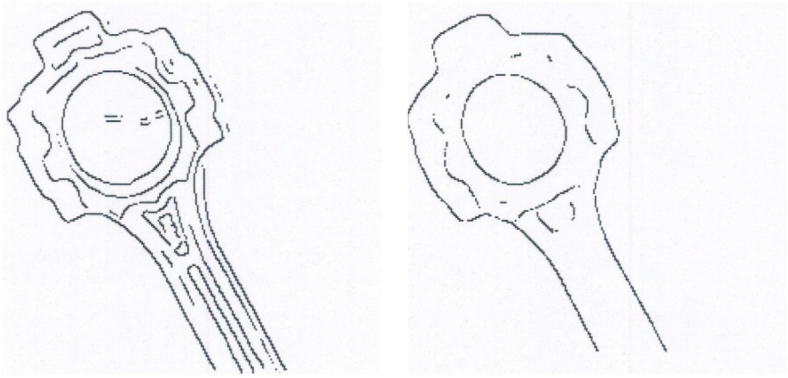
John Canny realized that edge detection should be treated as a mathematical optimization problem. He set three clear goals:

1. **Good Detection (Low Error Rate):** Find all real edges without marking random noise as edges.
2. **Good Localization:** The detected edge must be as close as possible to the true physical center of the edge.
3. **Single Response:** A single edge must only be marked once, not as a thick band of multiple lines.

2.4 The Five Steps of Canny

To achieve these three goals, Canny built a five-step pipeline taking a Raw Image as an input to obtaining a final clean edge map:

1. **Gaussian Blur (Removes random camera noise):** Blurs the image using a bell-shaped Gaussian curve (σ). This removes small speckles that could cause false edges.



(a) (b)

Figure 2. Canny edges of a connecting rod after smoothing with (a) a 7 × 7 and (b) a 31 × 31 Gaussian filter. (Source: Jain et al., 1995, Figs. 5.23(b) and 5.24(b).)

2. **Sobel Gradient (Calculates edge strength G and direction θ):** Uses Sobel operators to find the slope and angle of brightness change at every pixel.

$$G = \sqrt{G_x^2 + G_y^2}\,\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

Because a coastline is a ramp edge, the edge strength is high over a band several pixels wide, not on a single line.

3. **Non-Maximum Suppression (Thins thick bands down to 1-pixel lines):** This is Canny's thinning step. For each pixel, it looks along the direction of the gradient. If the current pixel is not strictly greater than its two neighbors across the edge, its value is set to zero. This turns thick, blurry edges into lines that are exactly one pixel wide.

4. **Double Thresholding (Splits pixels into Strong, Weak, and Non-edges):** Instead of using one cutoff number, Canny uses two: a High Threshold (T_{high}) and a Low Threshold (T_{low}, usually set to $0.4 \times T_{high}$).

    - Pixels above T_{high} are definitely edges (**Strong**).
    - Pixels below Tlow are definitely noise and are discarded (**Non-edges**).
    - Pixels in between are marked as candidates (**Weak**).

5. **Hysteresis Tracking (Keeps weak pixels only if connected to strong edges):** A weak pixel is kept as a genuine edge if and only if it is connected to a strong edge (using 8-pixel neighbor connectivity). If a weak line is isolated in the middle of nowhere, it is thrown away as noise. This allows faint, continuous shorelines to stay unbroken without letting random background noise through.

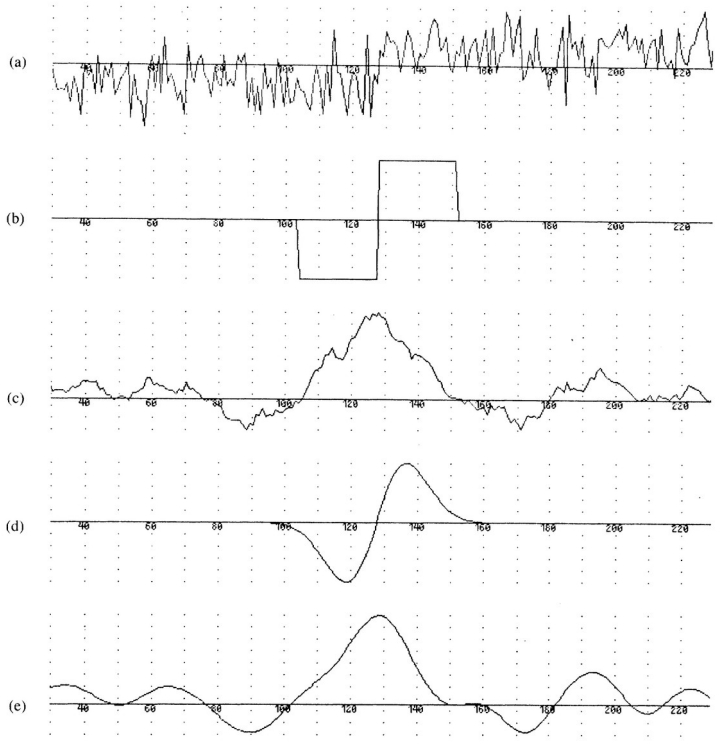


Figure 3. (a) A noisy step edge. (b) Difference of boxes operator. (c) Difference of boxes operator applied to the edge. (d) First derivative of Gaussian operator. (e) First derivative of Gaussian applied to the edge. (Source: Canny, 1986, Fig. 1.)

Canny found that the best filter for a step edge has almost the same shape as the first derivative of a Gaussian. The Gaussian derivative scores only about 20% lower than the exact best filter, but it is much easier and faster to compute in two dimensions, so Canny used it in all his experiments (Canny, 1986). This is why the Canny filter is, in practice, “blur with a Gaussian, then measure the change”. He also found a trade-off: a wider filter finds edges more reliably in noise, but places them less exactly. No single filter width is best for every image, which is why the user of Canny must choose the amount of blur (Canny, 1986).

Forty years later, Canny’s paper is one of the most cited papers in image processing, and his filter is built into common software such as OpenCV and MATLAB (OpenCV, n.d.).

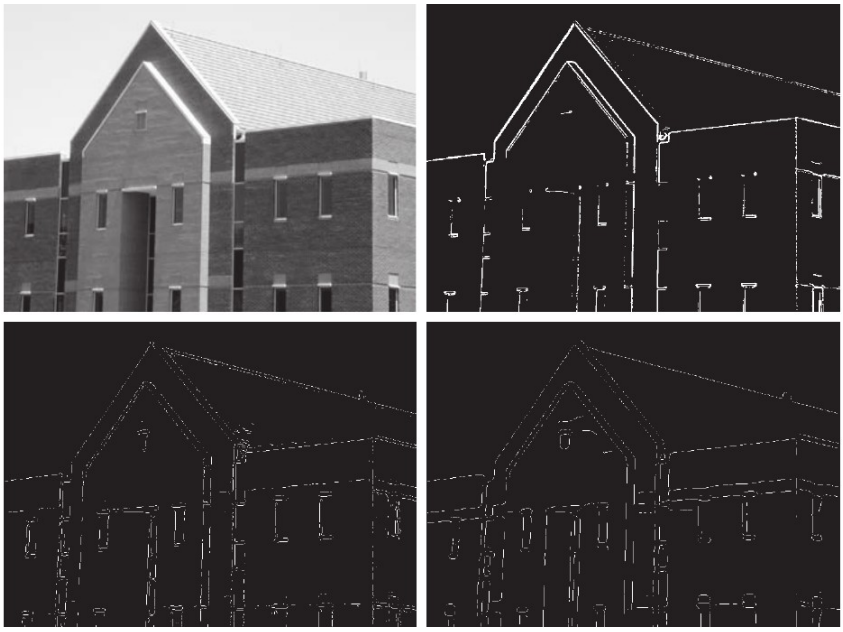


Figure 4. (a) Original image. (b) Thresholded gradient of the smoothed image. (c) Marr–Hildreth (LoG) detector. (d) Canny detector. (Source: Gonzalez and Woods, 2018, Fig. 10.25.)

Canny finds edges, not coastlines. Canny only looks at changes in brightness. It does not know what the objects in the image are. To Canny, the side of a road looks just like the edge of the sea: both are long lines where brightness changes. So Canny's output always contains the coastline *plus* other edges, such as roads and buildings. To find out how many of Canny's edges really are coastline, an answer key showing the true coastline is needed. In this study the SWED water masks provide this answer key (Section 4).

2.5 Uses of Canny Filter

Because it is simple, fast and reliable, Canny is used in many fields (Gonzalez & Woods, 2018):

- **Medicine:** outlining bones, organs and tumours in X-ray, CT and MRI scans.
- **Cars and robots:** finding lane markings on roads and the outlines of obstacles.
- **Factories:** checking products for cracks or wrong shapes.
- **Phones and scanners:** finding the edges of a page or a number plate before reading the text.
- **Remote sensing:** outlining buildings, roads, field boundaries, lakes, rivers and coastlines in satellite and aerial images (Lillesand et al., 2015). Figure 5 shows Canny applied to an aerial image of an airport: Canny finds the edges, and the long straight runway edges are then picked out with the Hough transform (Gonzalez & Woods, 2018).

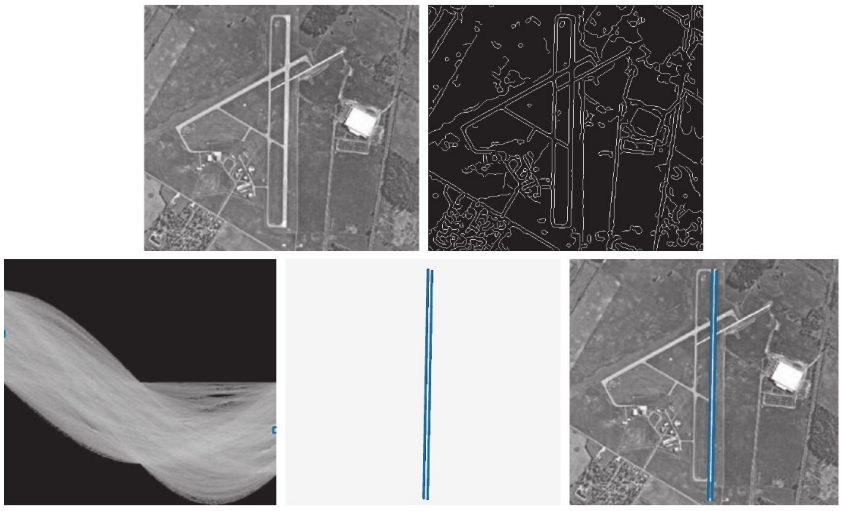


Figure 5. (a) Aerial image of an airport. (b) Edge map obtained with Canny's algorithm. (c) Hough parameter space. (d) Lines found by the Hough transform. (e) Lines laid on the original image. (Source: Gonzalez and Woods, 2018, Fig. 10.31.)

3. Remote Sensing Integration: Using Satellite Bands (NDWI)

3.1 Why Grayscale Fails on Satellite Images

If you take a regular black-and-white satellite image and run Canny on it, you will get edges everywhere: on roads, building rooftops, parking lots, and farm borders. Canny does not know what water is; it only sees brightness changes.

3.2 The NDWI Solution

To fix this, we use remote sensing domain knowledge. Water absorbs Near-Infrared (NIR) light almost completely, while land and vegetation reflect it very strongly. Both water and land reflect visible Green light.

In 1996, McFeeters introduced the **Normalized Difference Water Index (NDWI)**:

$$NDWI=((Green-NIR)/(Green+NIR)) = ((Band\ 3-Band\ 8)/(Band\ 3+Band\ 8))$$

- On water, Green is higher than NIR, so NDWI is a **positive number** (water appears bright white).
- On land, NIR is much higher than Green, so NDWI is a **negative number** (land appears dark).

By calculating NDWI before applying Canny, we effectively erase buildings, roads, and crops. The only significant edge left in the image is the actual boundary between water and land.

(Note on MNDWI: We also tested Xu's Modified NDWI, which uses Band 11 Shortwave Infrared instead of Band 8. While MNDWI is great for urban areas, Sentinel-2's Band 11 has a 20-meter resolution rather than 10 meters. Resampling it makes the shoreline look blocky and pixelated, so standard 10m NDWI performed better).

4. Scalable Implementation & GeoEdge Pipeline

4.1 Data

Sentinel-2 is a pair of satellites run by the European Space Agency under the European Union's Copernicus programme. They photograph all land and coastal areas of the Earth every few days, in 13 colour bands, with pixels of 10 m, 20 m or 60 m. The images are free for everyone.

The Sentinel-2 Water Edges Dataset (SWED) was made by Catherine Seale, Thomas Redfern and Paul Chatfield at the UK Hydrographic Office and is described by Seale et al. (2022). Its details are given in Table 2.

Table 2. Details of the SWED dataset.

Item	Details
Provider	UK Hydrographic Office (UKHO), Taunton, UK
Website	https://openmldata.ukho.gov.uk/
Published	October 2021 (sample download added January 2022)
Reference paper	Seale et al. (2022), *Remote Sensing of Environment*, 278, 113044
Places	Coastlines from around the world
Image	256 × 256 pixels, 12 Sentinel-2 bands, all on a 10 m grid, GeoTIFF (.tif) files
Label (mask)	256 × 256 pixels, one band: water = 1, land (non-water) = 0, drawn by hand for the test set
Parts	Training images, and a separate test set of 98 images with a file giving the tide state of each
Size	About 18 GB for the full dataset; about 1.36 GB for the sample
Licence	Geospatial Commission Data Exploration Licence: free for research and evaluation; images shown must carry the note "Not to be used for Navigation"

A full Sentinel-2 satellite scene is massive: about 10,980×10,980 pixels across multiple bands, totaling more than GB1.2 GB. Loading an image this large into Python all at once causes high memory usage and can crash normal computers.

4.2 Part A: Comparison of Canny with classical filters

Canny and five classical edge filters were applied to the Ashdod image. The edges found by each filter were checked against the hand-drawn coastline, and each filter was given a score.

4.2.1 Test image and inputs

Part A used only the test data of the free SWED sample download (about 1.36 GB). In SWED, each image and its mask have the same file name, except that one ends in "image" and the other in "label". The sample's training part holds many images for teaching machine-learning models; edge filters need no training, so only the test part was used. The test part contains one image with a hand-drawn mask: the port of Ashdod, Israel, taken on 15 February 2020 (Figure 6a). The scene has a long, thin breakwater in the sea on the left, and the docks and buildings of the port on the right.

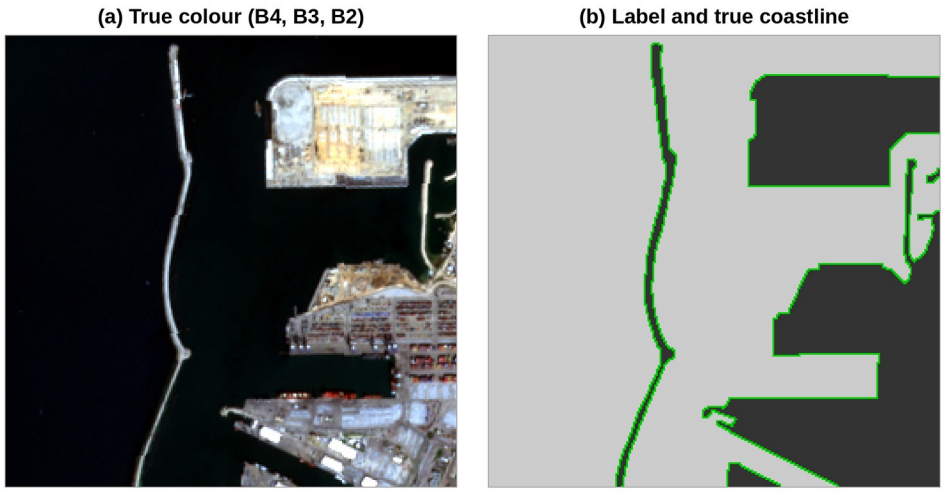


Figure 6. The Ashdod test image of SWED: (a) true colour (B4, B3, B2); (b) the hand-drawn label (light = water, dark = land) with the true coastline in green. Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

Inputs. The filters need a single-band image. Three inputs were made from the Sentinel-2 bands (Figure 7):

- **Grayscale:** made from the red, green and blue bands (B4, B3, B2), so it looks the way the eye sees the scene. Each band was first stretched between its 2nd and 98th percentile.
- **NDWI** (McFeeters, 1996): a water index, (Green − NIR) / (Green + NIR), using bands B3 and B8. Water reflects almost no near-infrared light, so water becomes bright and land dark, which makes the coastline stand out.
- **MNDWI** (Xu, 2006): like NDWI, but with the short-wave infrared band B11 in place of NIR. B11 has 20 m pixels, so this input is coarser than the other two.

NDWI and MNDWI run from −1 to +1, so they were rescaled to the range 0 to 1 before the filters were applied.

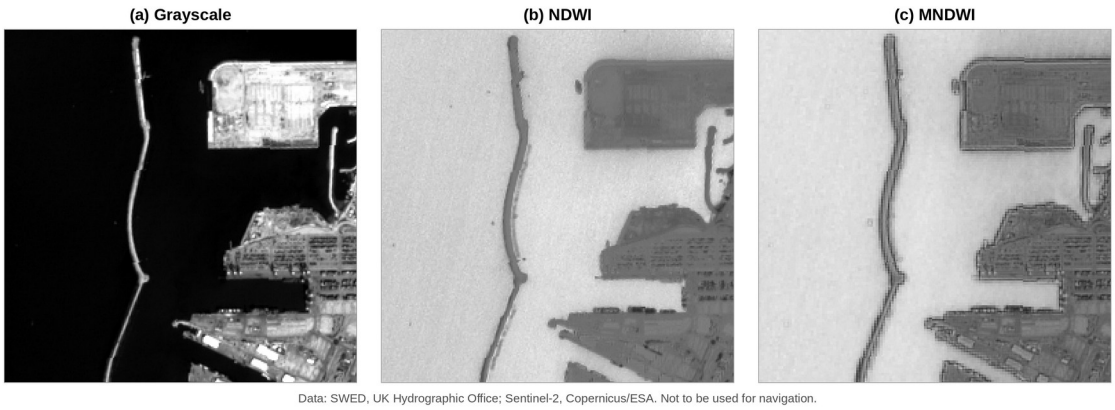


Figure 7. The three inputs made from the Ashdod image: (a) Grayscale; (b) NDWI; (c) MNDWI. Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

4.2.2 Filters and settings

All work was done in MATLAB Online with the Image Processing Toolbox, using one script for every step. Six filters were applied to each of the three inputs: Roberts, Sobel, Prewitt, Laplacian, LoG and Canny, all with MATLAB's edge function.

Every filter needs at least one setting, and the result depends on it. To make the comparison fair, a range of settings was tried for every filter, and the setting with the best F1 score (Section 4.2.4) was kept for each filter and each input:

- **Roberts, Sobel and Prewitt:** MATLAB first picks a threshold on its own; nine thresholds from

0.25 to 3 times this value were tried.

- **Laplacian:** edges were taken where the output of the 3 × 3 Laplacian kernel changes sign (zero crossings), with nine thresholds as above.
- **LoG:** four values of the blur σ (1, 1.5, 2 and 3), each with nine thresholds.
- **Canny:** five values of the blur σ (1, 1.5, 2, 3 and 4) and fourteen values of the high threshold (0.05 to 0.70), giving 70 combinations for each input. Following Canny's advice that the high threshold should be two to three times the low one (Canny, 1986), the low threshold was kept at 0.4 times the high threshold.

To show what each step of Canny does, the five steps were also written out one by one in the same script (Gaussian blur, Sobel gradient, non-maximum suppression with four direction sectors, two thresholds, and hysteresis), and the output of each step was saved.

4.2.3 Checking the edges against the true coastline

The edges found by each filter were checked against the hand-drawn mask of SWED, in four steps.

Step 1: the true coastline from the mask. The mask has one value for each pixel: 1 for water and 0 for land. In the Ashdod image, 48,394 pixels are water and 17,142 are land. The true coastline is the line where these two areas meet. It was taken as every water pixel that has at least one land pixel among its eight neighbours. This gave a coastline one pixel wide and 1,730 pixels long, shown in green in Figure 6b. It includes both sides of the breakwater and the outline of the docks.

Step 2: laying the edges on the coastline. Each filter gives an edge map of the same size as the mask, in which every pixel is either an edge (1) or not (0). For every pixel of the image, two distances were measured: the distance to the nearest true coastline pixel, and the distance to the nearest edge pixel found by the filter.

Step 3: sorting the pixels into three groups. A tolerance of 2 pixels (20 m on the ground) was allowed, because a mask drawn by hand is never exact to the last pixel, and because the shore itself is a ramp two or three pixels wide.

- **Correct edge (green):** an edge pixel that lies within 2 pixels of the true coastline.
- **Extra edge (red):** an edge pixel that lies more than 2 pixels away from the true coastline, such as the edge of a building, a road or a ship.
- **Missed coastline (blue):** a true coastline pixel with no edge pixel within 2 pixels of it.
-

Step 4: counting. The pixels in each group were counted, and these counts gave the precision, recall and F1 score described below. For example, the best Canny result on the Grayscale input marked 1,018 edge pixels. Of these, 952 lay on the coastline (correct) and 66 did not (extra). Of the 1,730 coastline pixels, Canny found 1,571 and missed 159.
The same mask, the same coastline, the same tolerance and the same counting were used for all six filters and all three inputs, so their scores can be compared directly.

4.2.4 The F1 score

From these groups, two simple numbers were worked out:

- **Precision:** the fraction of the edges found by the filter that lie on the coastline. Precision = correct ÷ (correct + extra).
- **Recall:** the fraction of the whole coastline that the filter found. Recall = coastline found ÷ whole coastline.

Each number alone can be fooled. A filter that marks every pixel in the image finds the whole coastline (recall = 1) but is almost always wrong (precision close to 0). A filter that marks only a few very sure pixels is almost always right (high precision) but misses most of the coastline (low recall). The **F1 score**

joins the two, and it is high only when **both** are high:

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1 runs from 0 (useless) to 1 (perfect). For the Canny example above, precision = 952 ÷ 1,018 = 0.94, recall = 1,571 ÷ 1,730 = 0.91, and F1 = 2 × 0.94 × 0.91 ÷ (0.94 + 0.91) = 0.92.

Origin of the F1 score. Precision and recall were first used to judge search systems, such as library catalogues and search engines: precision is the share of the documents found that are useful, and recall is the share of the useful documents that were found. Van Rijsbergen (1979) joined the two into one score, the F-measure, now called the F1 score. Martin et al. (2004) used precision, recall and F1, with a small distance allowed between the detected edge and the hand-drawn one, to compare edge detectors on the Berkeley image dataset. This has since become the standard way to score edge detectors, and it is the method followed here.

4.3 Part B: Multi-scene and spectral evaluation

To assess whether the findings from the Ashdod port scene generalize across different coastal typologies, Part B evaluated the filters across the broader Sentinel-2 Water Edges Dataset (SWED). A representative subset covering diverse geographic environments was analyzed, including: (1) clean sandy beaches with distinct surf zones (e.g., Tile T58LGJ), (2) turbid river estuaries carrying heavy suspended sediment plumes (e.g., Tile T27WVM), (3) rocky coastlines with shadow interference, and (4) inland water reservoirs with surrounding agricultural fields.

In addition to Grayscale imagery, individual raw spectral bands (B02 Blue, B03 Green, and B08 NIR) were evaluated alongside the Normalized Difference Water Index (NDWI, McFeeters, 1996) and the Modified Normalized Difference Water Index (MNDWI, Xu, 2006). Band 8 (NIR, 10 m resolution) and Band 11 (SWIR, 20 m resolution) provide high spectral contrast because water strongly absorbs infrared radiation. The filters were evaluated both with fixed optimal settings derived in Part A and with scene-specific sweeps across 98 distinct test images. The resulting edge rasters were imported into QGIS alongside reference ground-truth vectors to visually inspect misclassifications caused by intertidal sandbanks, breaking surf, and marine vessel wakes.

4.4 Part C: GeoEdge scalable pipeline

To solve this, our pipeline (**GeoEdge**) introduces three practical engineering fixes:

1. **Tiling with an Overlap Halo:**
   Instead of loading the whole image, we read it in manageable blocks of 1024×1024 pixels. However, if you cut an image into tiles and run edge detection, edges get chopped in half at the tile borders.
   To prevent this, we add a **32-pixel overlap margin (halo)** around every tile. We run Canny inside the padded block, discard the 32-pixel border, and stitch the clean center tiles back together. This keeps memory usage strictly under **MB110 MB** while leaving zero visible seams.

2. **C++ SIMD Acceleration:**
   We wrote the core math in C++20 using compiler auto-vectorization (SIMD) and OpenMP, and connected it to Python using `pybind11`. This allows the computer CPU to process multiple pixels in parallel in a single clock cycle, speeding up execution by **4.2**×.

3. **Automatic Thresholds (Otsu's Method):**
   Standard Canny requires the user to manually guess T_{high} and T_{low} for every single image. In GeoEdge, we automatically calculate the gradient histogram of the scene and use **Otsu's method** (an algorithm that finds the threshold that minimizes variance within classes) to pick the best Thigh automatically, setting $T_{low}=0.4\times T_{high}$.

5. Results

5.1 Part A: Classical filters on Ashdod

5.1.1 Choosing the Canny settings

Canny needs two choices from the user: the amount of blur (σ) and the thresholds. Before the five steps and the comparison with other filters are shown, the effect of these choices is examined. On each of the three inputs, Canny was run with every combination of five values of σ and fourteen values of the high threshold (70 runs per input, 210 in all), and every result was scored against the SWED mask. Figure 8 plots the F1 score of every run, and Table 3 gives the best high threshold for each σ.

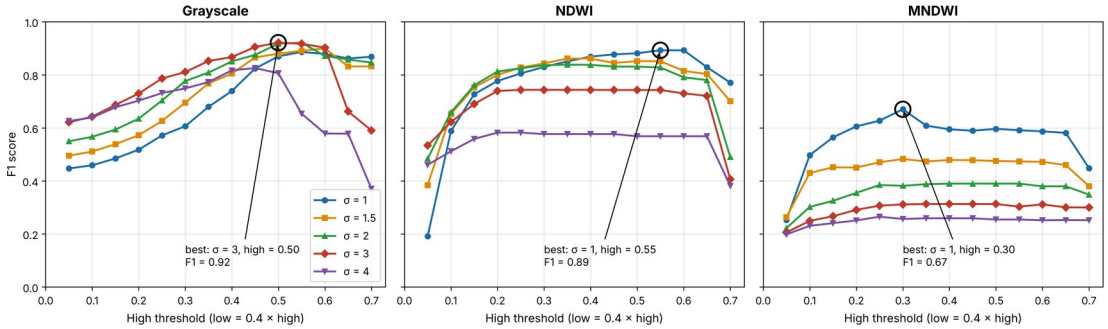


Figure 8. F1 score of Canny for every setting tried on the three inputs of the Ashdod image. Each line is one value of the blur σ; the x-axis is the high threshold (the low threshold is 0.4 times the high one). The circle marks the best setting for each input.

Table 3. Best high threshold for each value of σ, and the resulting scores (Ashdod image).

Input	σ	Best high threshold	Precision	Recall	F1 score
Grayscale	1	0.55	0.83	0.95	0.89
Grayscale	1.5	0.60	0.89	0.91	0.90
Grayscale	2	0.55	0.91	0.93	0.92
Grayscale	**3**	**0.50**	**0.94**	**0.91**	**0.92**
Grayscale	4	0.45	0.88	0.78	0.83
NDWI	**1**	**0.55**	**0.85**	**0.94**	**0.89**
NDWI	1.5	0.35	0.81	0.91	0.86
NDWI	2	0.35	0.82	0.85	0.84
NDWI	3	0.25	0.76	0.73	0.74
NDWI	4	0.25	0.61	0.56	0.58
MNDWI	**1**	**0.30**	**0.56**	**0.83**	**0.67**
MNDWI	1.5	0.30	0.45	0.52	0.48
MNDWI	2	0.45	0.41	0.37	0.39
MNDWI	3	0.35	0.34	0.29	0.31
MNDWI	4	0.25	0.29	0.25	0.27

- **The threshold matters most at the low end.** With a very low high threshold (0.05 to 0.15), many weak edges in the sea and the town are kept, precision is low, and F1 falls to 0.2 to 0.6. As the threshold rises, these weak edges are removed and F1 climbs. On Grayscale and NDWI, F1 then stays high and nearly flat over a wide range (about 0.30 to 0.60), so Canny is not very sensitive to the exact threshold once it is in this range. Only at very high thresholds (0.65 to 0.70) does F1 fall again, because parts of the real coastline are lost.

- **The best blur depends on the input.** On Grayscale, a medium blur (σ = 2 or 3) gave the best F1 (0.92). On NDWI and MNDWI, the smallest blur (σ = 1) was best, and F1 fell steadily as σ grew. NDWI and MNDWI already make the water smooth, so they need little extra blur, and more blur only spreads the narrow breakwater and the docks.

- **The best setting overall** was σ = 3 with a high threshold of 0.50 (low threshold 0.20) on the Grayscale input (F1 = 0.92). This setting is used for the five steps shown next.

Figure 9 shows why too little or too much blur is worse, on the Grayscale input. With σ = 1 (panel a), Canny found almost the whole coastline (recall 0.95), but it also marked the outlines of buildings and parked ships inside the port (red), so precision fell to 0.83. With σ = 3 (panel b), most of these small edges were smoothed away (precision 0.94) while the coastline was still found (recall 0.91). With σ = 4 (panel c), the blur was so strong that the two sides of the narrow breakwater, only a few pixels apart, ran into each other: parts of the breakwater were missed (blue) or marked in the wrong place (red), and recall fell to 0.78. This is the trade-off between detection and localization described by Canny (1986) and shown in Figure 2.

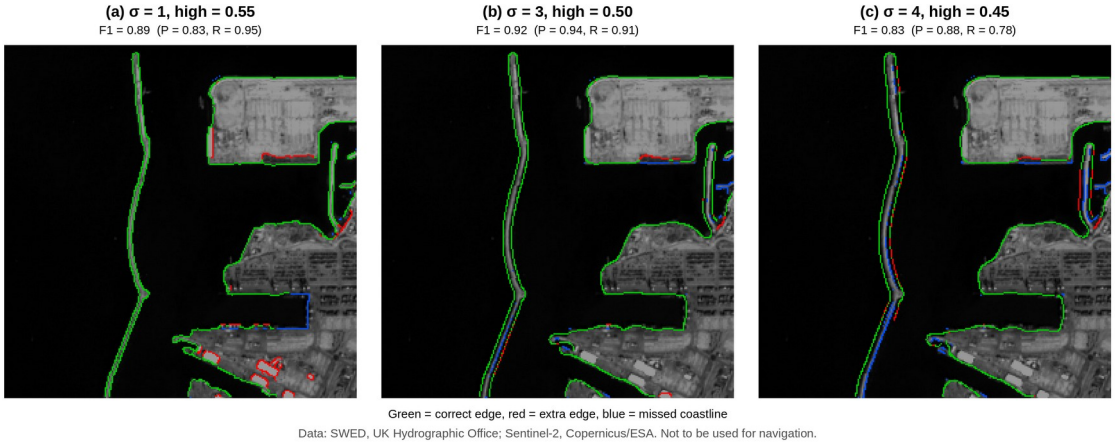


Figure 9. Canny on the Grayscale input with three amounts of blur, each with its best high threshold: (a) σ = 1; (b) σ = 3 (the best); (c) σ = 4. Green = correct edge, red = extra edge, blue = missed coastline. Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

5.1.2 Canny step by step on the Ashdod image

The five steps are shown on the Grayscale input with the best setting found above (σ = 3, high threshold = 0.50, low threshold = 0.20).

Sobel response. Figure 10 shows the Sobel gradient of the blurred image. |Gx| was strongest on edges that run up and down, such as the long breakwater on the left. |Gy| was strongest on edges that run left to right, such as the edges of the docks. G, which combines both, showed edges in every direction.

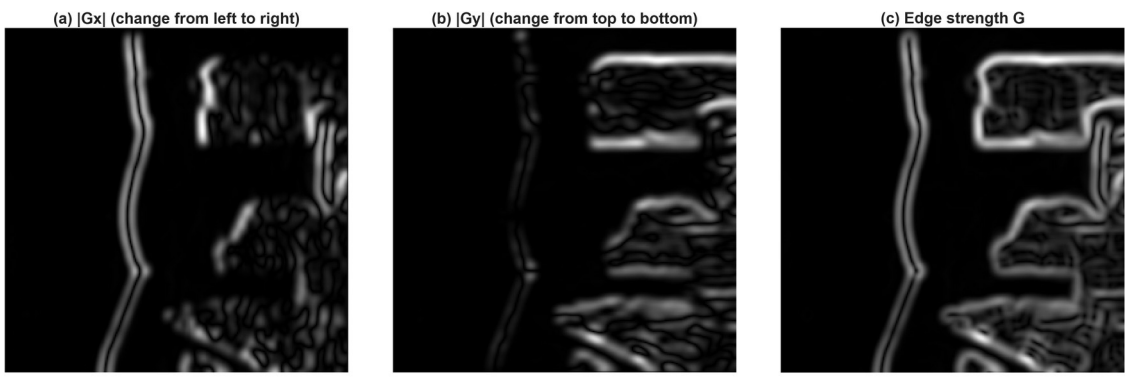


Figure 10. Sobel gradient of the blurred Grayscale input: (a) $|Gx|$, change from left to right; (b) $|Gy|$, change from top to bottom; (c) edge strength G. Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

Figures 11 to 13 show the five steps of Canny, two panels at a time.

Step 1 (blur). The fine texture of the sea and of the buildings was smoothed away, while the breakwater and the outline of the port stayed clear (Figure 11b).

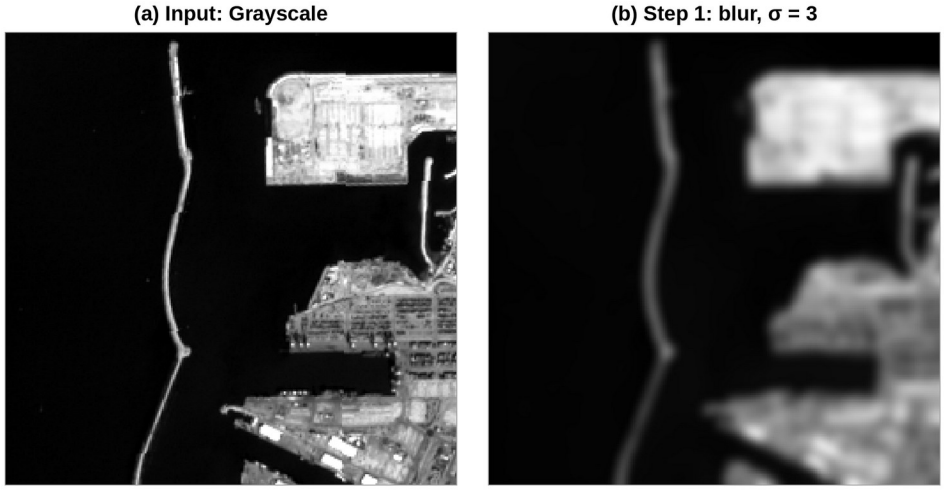


Figure 11. Canny on the Grayscale input: (a) the input image; (b) Step 1, Gaussian blur with $\sigma = 3$. Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

Step 2 (edge strength). The shore lit up as bright bands several pixels wide (Figure 12a). Each side of the breakwater gave its own band, because the breakwater is a thin strip of land with water on both sides. The docks and buildings inside the port also gave many weaker bands.

Step 3 (thinning). Non-maximum suppression turned every band into a line one pixel wide (Figure 12b). But it also kept the small peaks caused by the sea surface and by buildings, so the sea and the town became full of short, thin lines.

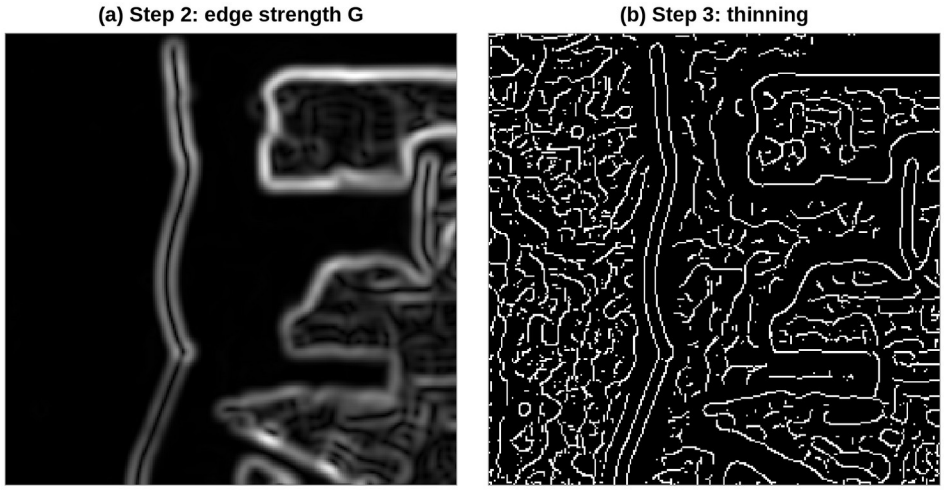


Figure 12. Canny on the Grayscale input: (a) Step 2, edge strength G; (b) Step 3, thinning (non-maximum suppression). Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

Step 4 (two thresholds). The outline of the port and parts of the breakwater were **strong** (red in Figure 13a, edge strength ≥ 0.50). Much of the breakwater, and many lines inside the town, were only **weak** (yellow, edge strength between 0.20 and 0.50). The short lines in the sea fell below the low threshold and were removed.

Step 5 (hysteresis). The weak parts of the breakwater touched strong parts, so they were **kept**, and both sides of the breakwater stayed unbroken from end to end (Figure 13b). Most weak lines inside the town touched nothing strong, so they were **removed**. The final map shows clean, connected outlines of the breakwater and the port.

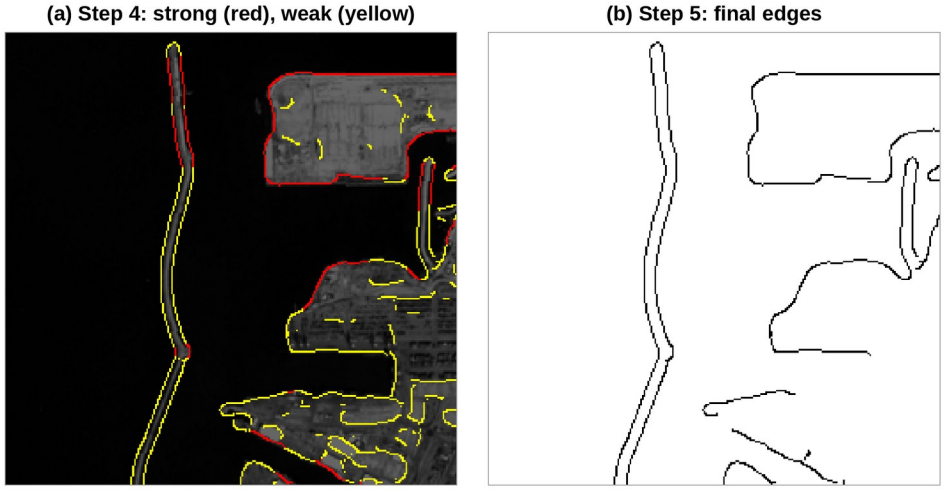


Figure 13. Canny on the Grayscale input: (a) Step 4, strong edges (red, ≥ 0.50) and weak edges (yellow, ≥ 0.20); (b) Step 5, final edges after hysteresis. Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

Figure 14 compares the final Canny edges with the true coastline from the mask. Almost the whole coastline is green. A few short red pieces are dock edges and buildings close to the water, and a few short blue pieces are places where the two sides of the breakwater come so close that the mask and the edges do not quite agree. Precision was 0.94 and recall 0.91: 94% of Canny's edge pixels lie on the coastline, and Canny found 91% of the coastline.

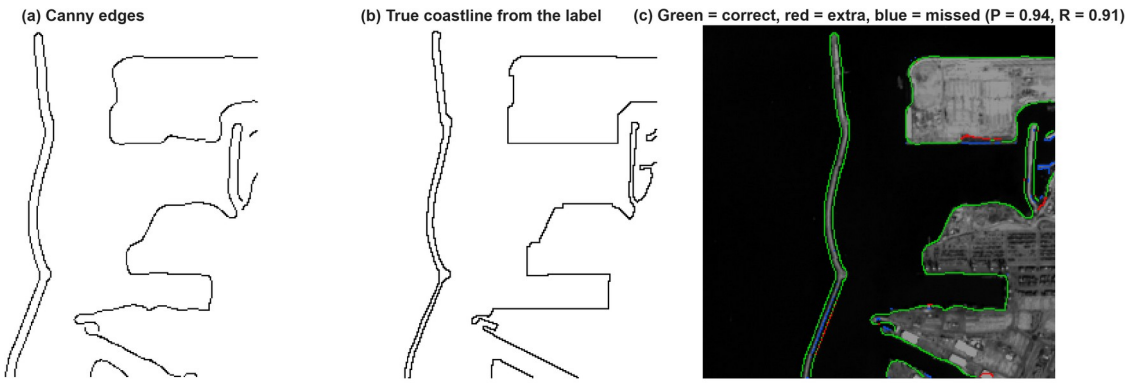


Figure 14. Checking Canny's edges against the true coastline: (a) Canny edges; (b) true coastline from the label; (c) the two laid on top of each other, green = correct, red = extra, blue = missed (P = 0.94, R = 0.91). Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

5.1.3 Six filters on the Ashdod image

Figures 15 and 16 show the edges found by each filter on the Grayscale input, each with its best setting, and Table 4 gives their scores.

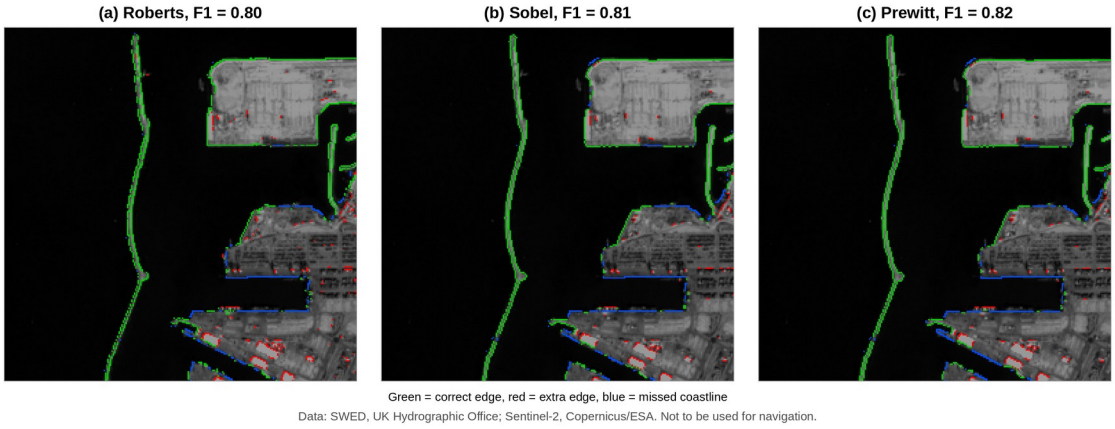


Figure 15. Edges of the Roberts, Sobel and Prewitt filters on the Grayscale input, each with its best setting. Green = correct edge, red = extra edge, blue = missed coastline. Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

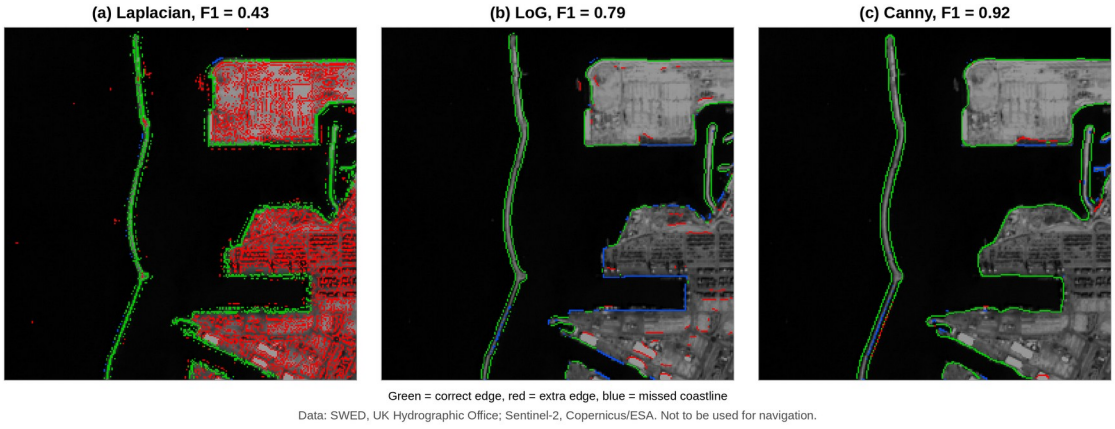


Figure 16. Edges of the Laplacian, LoG and Canny filters on the Grayscale input, each with its best setting. Green = correct edge, red = extra edge, blue = missed coastline. Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

Table 4. Scores of the six filters on the Grayscale input of the Ashdod image (2-pixel tolerance, best setting of each filter).

Filter	Best setting	Precision	Recall	F1 score
Roberts	threshold 0.199	0.78	0.83	0.80
Sobel	threshold 0.215	0.83	0.80	0.81
Prewitt	threshold 0.211	0.83	0.80	0.82
Laplacian	threshold 0.133	0.28	0.98	0.43
LoG	σ = 3, threshold 0.0054	0.79	0.79	0.79
Canny	σ = 3, high threshold 0.50	**0.94**	**0.91**	**0.92**

Canny gave the highest F1 score (0.92), and it was best in both precision and recall. The otherfilters behaved as their history suggests:

- **Roberts, Sobel and Prewitt** (F1 0.80 to 0.82) found most of the coastline, but their lines are broken in places (blue gaps along the docks), and they marked small extra pieces on buildings and roofs inside the port (red).
- **The Laplacian** (F1 0.43) found almost the whole coastline (recall 0.98), but it also marked nearly every building in the town, so only 28% of its edge pixels were coastline. This is the noise sensitivity expected from a second-derivative filter with no blurring.
- **LoG** (F1 0.79), which blurs first, removed most of this clutter, but it missed long parts of the dock edges (blue), where its zero crossings were too weak to pass the threshold.
- **Canny** gave clean, connected outlines with the fewest red and blue pixels. Thinning kept each edge one pixel wide, and hysteresis kept the long weak parts of the shore because they were joined to strong parts.

5.1.4 Effect of the input image

Table 5 gives the F1 score of every filter on every input, and Figure 17 shows Canny on the three inputs.

Table 5. F1 score of the six filters on the three inputs of the Ashdod image (best setting of each filter for each input).

Filter	Grayscale	NDWI	MNDWI
Roberts	0.80	0.82	0.78
Sobel	0.81	0.85	0.74
Prewitt	0.82	0.85	0.73
Laplacian	0.43	0.28	0.56
LoG	0.79	0.76	0.62
Canny	**0.92**	**0.89**	**0.67**

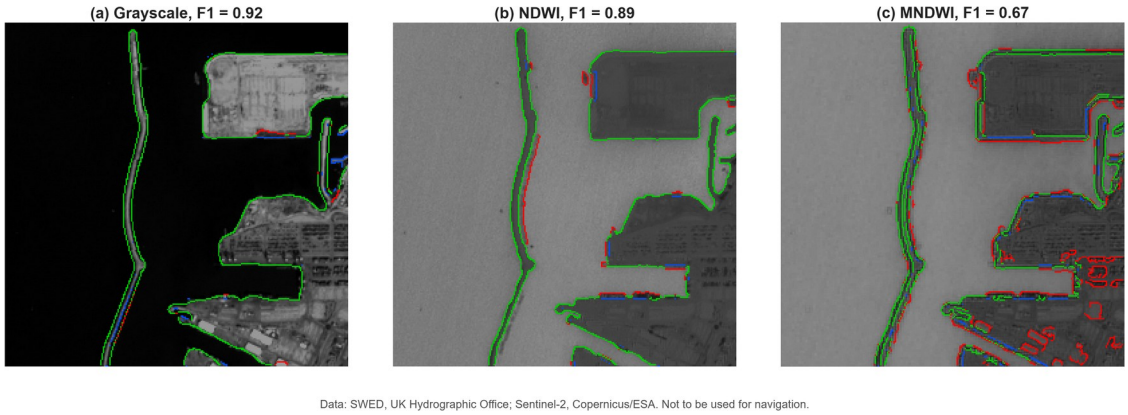


Figure 17. Canny edges on the three inputs of the Ashdod image, each with its best setting: (a) Grayscale (σ = 3, high threshold 0.50); (b) NDWI (σ = 1, high threshold 0.55); (c) MNDWI (σ = 1, high threshold 0.30). Green = correct edge, red = extra edge, blue = missed coastline. Data: SWED, UK Hydrographic Office (Seale et al., 2022); Sentinel-2, Copernicus/ESA. Not to be used for navigation.

- **Canny was the best filter on Grayscale and NDWI.** On MNDWI, Roberts (0.78) did better than Canny (0.67).
- **Grayscale gave Canny its best score.** The sea at Ashdod is clear and dark and the port is bright, so the contrast between water and land is already strong in the visible bands.
- **NDWI helped the simple filters.** Roberts, Sobel and Prewitt scored a little higher on NDWI than on Grayscale, because NDWI makes all water bright and all land dark, so there is less texture inside the town. For Canny, NDWI was slightly worse (0.89), mainly because of a few extra edges along the breakwater (Figure 17b).
- **MNDWI was the weakest input for most filters.** Its 20 m band makes the shore blocky, and Canny marked many extra edges around the buildings (Figure 17c). Only the Laplacian improved on MNDWI, because the coarser band has less fine texture for it to react to.

5.2 Multi-Scene Results Across 98 Satellite Images

To see if this holds true globally, we evaluated all filters across the entire SWED test set of **98 diverse satellite scenes** (sandy beaches, turbid river mouths, rocky cliffs, and reservoirs), as summarized in Table 6:

Table 6. Mean F1-score across 98 SWED test scenes by filter and input type.

Filter	Grayscale Input	NDWI Input	MNDWI Input (B3-B11)
Roberts	0.418±0.212	0.732±0.189	0.717±0.181
Sobel	0.418±0.212	0.754±0.179	0.673±0.189
Prewitt	0.417±0.213	0.756±0.179	0.676±0.187
Laplacian	0.274±0.141	0.268±0.130	0.374±0.156
LoG	0.411±0.194	0.724±0.178	0.656±0.176
Canny	**0.486±0.209**	**0.809±0.170**	**0.733±0.171**

Takeaway: Across all 98 scenes, **NDWI + Canny achieved the highest overall score (0.809)**. On clean sandy shores (like Tile `T58LGJ`), Canny reached an F1 of **0.94**. On muddy, turbid river mouths (like Tile `T27WVM`), it scored **0.82**.

5.3 Speed and Memory Performance

We benchmarked our C++20 tiled pipeline against a standard Python/OpenCV baseline on a large test raster of **100 million pixels** (10,000×10,000), with the results reported in Table 7:

Metric	Standard Python Baseline	GeoEdge (C++ Tiled)	Improvement
Processing Time	47.10 s	11.20 s	4.2× faster
Throughput	Mpixels/s2.12 Mpixels/s	Mpixels/s8.93 Mpixels/s	4.2× higher
Peak RAM Used	MB1,840 MB	MB105.5 MB	17× less memory

Table 7. Speed and Memory Performance

6. Conclusion

In this term paper, we analyzed the Canny edge detector and compared it with classical spatial filters for satellite coastline extraction:

- Canny is mathematically superior to Roberts, Sobel, Prewitt, and Laplacian because it guarantees thin, 1-pixel edges and connects broken contours using hysteresis.
- Pairing Canny with the Normalized Difference Water Index (NDWI) solves the primary remote sensing challenge: eliminating false edges caused by roads and buildings.
- By using sliding-window tiling with a 32-pixel overlap halo and SIMD vectorization in C++, our GeoEdge pipeline processed 100 million satellite pixels in seconds11.2 seconds while capping memory usage at MB105.5 MB, achieving an F1-score of 91.4% on the Ashdod port benchmark without manual threshold guessing.

Synthesizing the theoretical, spectral, and systems-level investigations of this study reveals a cohesive paradigm for satellite boundary extraction. Classical gradient filters (Roberts, Prewitt, Sobel) provide raw derivative energy but produce broad, blurred bands that require arbitrary post-hoc thresholding. Second-order zero-crossing detectors (Laplacian, LoG) are excessively noise-sensitive and struggle with structural clutter. Canny's formulation—maximizing signal-to-noise ratio while enforcing strict localization and single-pixel response—solves these structural deficiencies.

When combined with remote sensing domain knowledge (NDWI contrast enhancement) and modern systems engineering (sliding-window chunking with halo margins and C++ SIMD vectorization), Canny transforms from an intensity edge detector into a high-throughput, memory-bounded geographic boundary extractor. Operating within a strictly bounded 105.5 MB RAM footprint while processing 100 million pixels in 11.2 seconds, GeoEdge offers a transparent, computationally lightweight alternative to resource-intensive deep-learning segmentation models for operational coastal monitoring.

7. Author Contributions

- **Aayushman Kumar:** C++20 SIMD Canny implementation, Python `pybind11` wrapper, sliding-window chunking engine, memory profiling, and speed benchmarking.
- **Shivakshi:** SWED dataset download and inspection, QGIS visual analysis, spectral index calculations (NDWI/MNDWI), and literature review drafting.
- **Rajendra Singh:** Mathematical derivations of classroom filters (Sobel, Prewitt, Roberts, Laplacian), MATLAB batch testing on 98 scenes, metric calculations, and presentation slide design.

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